- if $a = b$, then $\frac{a}{c} = \frac{b}{c}$, $c \neq 0$. Division Property of Equality

- **The Multiplication Property of Equality**
  - For any numbers ab and c, $a = b$, then $ac = bc$.
  - If you multiply both sides of an equation by the same quantity, you still have equality.

Exercises

Review Exercises

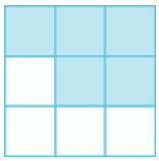
Visualize Fractions

In the following exercises, name the fraction of each figure that is shaded.

574.

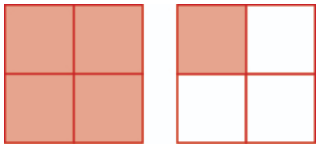


575.

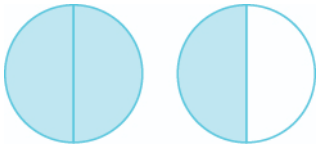


In the following exercises, name the improper fractions. Then write each improper fraction as a mixed number.

576.



577.



In the following exercises, convert the improper fraction to a mixed number.

578. $\frac{58}{15}$

579. $\frac{63}{11}$

In the following exercises, convert the mixed number to an improper fraction.

580. $12\frac{1}{4}$

581. $9\frac{4}{5}$

582. Find three fractions equivalent to $\frac{2}{5}$. Show your work, using figures or algebra.

583. Find three fractions equivalent to $-\frac{4}{3}$. Show your work, using figures or algebra.

In the following exercises, locate the numbers on a number line.

584. $\frac{5}{8}, \frac{4}{3}, 3\frac{3}{4}, 4$

585. $\frac{1}{4}, -\frac{1}{4}, 1\frac{1}{3}, -1\frac{1}{3}, \frac{7}{2}, -\frac{7}{2}$

In the following exercises, order each pair of numbers, using $<$ or $>$.

586. -1 ___ $-\frac{2}{5}$

587. $-2\frac{1}{2}$ ___ -3

Multiply and Divide Fractions

In the following exercises, simplify.

588. $-\frac{63}{84}$

589. $-\frac{90}{120}$

590. $-\frac{14a}{14b}$

591. $-\frac{8x}{8y}$

In the following exercises, multiply.

592. $\frac{2}{5} \cdot \frac{8}{13}$

593. $-\frac{1}{3} \cdot \frac{12}{7}$

594. $\frac{2}{9} \cdot \left(-\frac{45}{32}\right)$

595. $6m \cdot \frac{4}{11}$

596. $-\frac{1}{4}(-32)$

597. $\frac{16}{5} \cdot \frac{15}{8}$

In the following exercises, find the reciprocal.

598. $\frac{2}{9}$

599. $\frac{15}{4}$

600. 3

601. $-\frac{1}{4}$

602. Fill in the chart.

	Opposite	Absolute Value	Reciprocal
$-\frac{5}{13}$			
$\frac{3}{10}$			
$\frac{9}{4}$			
-12			

In the following exercises, divide.

603. $\frac{2}{3} \div \frac{1}{6}$

604. $\left(-\frac{3x}{5}\right) \div \left(-\frac{2y}{3}\right)$

605. $\frac{4}{5} \div 3$

606. $8 \div \frac{8}{3}$

607. $\frac{5}{18} \div \left(-\frac{b}{9}\right)$

Multiply and Divide Mixed Numbers and Complex Fractions

In the following exercises, perform the indicated operation.

608. $3\frac{1}{5} \cdot 1\frac{7}{8}$

609. $-5\frac{7}{12} \cdot 4\frac{4}{11}$

610. $8 \div 2\frac{2}{3}$

611. $8\frac{2}{3} \div 1\frac{1}{12}$

In the following exercises, translate the English phrase into an algebraic expression.

612. the quotient of 8 and y

613. the quotient of V and the difference of h and 6

In the following exercises, simplify the complex fraction

614. $\dfrac{\frac{5}{8}}{\frac{4}{5}}$

615. $\dfrac{\frac{8}{9}}{-4}$

616. $\dfrac{\frac{n}{4}}{\frac{3}{8}}$

617. $\dfrac{-1\frac{5}{6}}{-\frac{1}{12}}$

In the following exercises, simplify.

618. $\frac{5+16}{5}$

619. $\frac{8 \cdot 4 - 5^2}{3 \cdot 12}$

620. $\frac{8 \cdot 7 + 5(8-10)}{9 \cdot 3 - 6 \cdot 4}$

Add and Subtract Fractions with Common Denominators

In the following exercises, add.

621. $\frac{3}{8}+\frac{2}{8}$

622. $\frac{4}{5}+\frac{1}{5}$

623. $\frac{2}{5}+\frac{1}{5}$

624. $\frac{15}{32}+\frac{9}{32}$

625. $\frac{x}{10}+\frac{7}{10}$

In the following exercises, subtract.

626. $\frac{8}{11}-\frac{6}{11}$

627. $\frac{11}{12}-\frac{5}{12}$

628. $\frac{4}{5}-\frac{y}{5}$

629. $-\frac{31}{30}-\frac{7}{30}$

630. $\frac{3}{2}-\left(\frac{3}{2}\right)$

631. $\frac{11}{15}-\frac{5}{15}-\left(-\frac{2}{15}\right)$

Add and Subtract Fractions with Different Denominators

In the following exercises, find the least common denominator.

632. $\frac{1}{3}$ and $\frac{1}{12}$

633. $\frac{1}{3}$ and $\frac{4}{5}$

634. $\frac{8}{15}$ and $\frac{11}{20}$

635. $\frac{3}{4}$, $\frac{1}{6}$,and$\frac{5}{10}$

In the following exercises, change to equivalent fractions using the given LCD.

636. $\frac{1}{3}$ and $\frac{1}{5}$, LCD $= 15$

637. $\frac{3}{8}$ and $\frac{5}{6}$, LCD $= 24$

638. $-\frac{9}{16}$ and $\frac{5}{12}$, LCD $= 48$

639. $\frac{1}{3}$, $\frac{3}{4}$ and $\frac{4}{5}$, LCD $= 60$

In the following exercises, perform the indicated operations and simplify.

640. $\frac{1}{5}+\frac{2}{3}$

641. $\frac{11}{12}-\frac{2}{3}$

642. $-\frac{9}{10}-\frac{3}{4}$

643. $-\frac{11}{36}-\frac{11}{20}$

644. $-\frac{22}{25}+\frac{9}{40}$

645. $\frac{y}{10}-\frac{1}{3}$

646. $\frac{2}{5}+\left(-\frac{5}{9}\right)$

647. $\frac{4}{11}\div\frac{2}{7d}$

648. $\frac{2}{5}+\left(-\frac{3n}{8}\right)\left(-\frac{2}{9n}\right)$

649. $\dfrac{\left(\frac{2}{3}\right)^2}{\left(\frac{5}{8}\right)^2}$

650. $\left(\frac{11}{12}+\frac{3}{8}\right)\div\left(\frac{5}{6}-\frac{1}{10}\right)$

In the following exercises, evaluate.

651. $y-\frac{4}{5}$ when
ⓐ $y=-\frac{4}{5}$ ⓑ $y=\frac{1}{4}$

652. $6mn^2$ when $m=\frac{3}{4}$ and $n=-\frac{1}{3}$

Add and Subtract Mixed Numbers

In the following exercises, perform the indicated operation.

653. $4\frac{1}{3}+9\frac{1}{3}$

654. $6\frac{2}{5}+7\frac{3}{5}$

655. $5\frac{8}{11}+2\frac{4}{11}$

656. $3\frac{5}{8}+3\frac{7}{8}$

657. $9\frac{13}{20}-4\frac{11}{20}$

658. $2\frac{3}{10}-1\frac{9}{10}$

659. $2\frac{11}{12}-1\frac{7}{12}$

660. $8\frac{6}{11}-2\frac{9}{11}$

Solve Equations with Fractions

In the following exercises, determine whether the each number is a solution of the given equation.

661. $x-\frac{1}{2}=\frac{1}{6}$:
ⓐ $x=1$ ⓑ $x=\frac{2}{3}$
ⓒ $x=-\frac{1}{3}$

662. $y+\frac{3}{5}=\frac{5}{9}$:
ⓐ $y=\frac{1}{2}$ ⓑ $y=\frac{52}{45}$
ⓒ $y=-\frac{2}{45}$

In the following exercises, solve the equation.

663. $n + \frac{9}{11} = \frac{4}{11}$

664. $x - \frac{1}{6} = \frac{7}{6}$

665. $h - \left(-\frac{7}{8}\right) = -\frac{2}{5}$

666. $\frac{x}{5} = -10$

667. $-z = 23$

In the following exercises, translate and solve.

668. The sum of two-thirds and n is $-\frac{3}{5}$.

669. The difference of q and one-tenth is $\frac{1}{2}$.

670. The quotient of p and -4 is -8.

671. Three-eighths of y is 24.

Practice Test

Convert the improper fraction to a mixed number.

672. $\frac{19}{5}$

Convert the mixed number to an improper fraction.

673. $3\frac{2}{7}$

Locate the numbers on a number line.

674. $\frac{1}{2}, 1\frac{2}{3}, -2\frac{3}{4}$, and $\frac{9}{4}$

In the following exercises, simplify.

675. $\frac{5}{20}$

676. $\frac{18r}{27s}$

677. $\frac{1}{3} \cdot \frac{3}{4}$

678. $\frac{3}{5} \cdot 15$

679. $-36u\left(-\frac{4}{9}\right)$

680. $-5\frac{7}{12} \cdot 4\frac{4}{11}$

681. $-\frac{5}{6} \div \frac{5}{12}$

682. $\frac{7}{11} \div \left(-\frac{7}{11}\right)$

683. $\frac{9a}{10} \div \frac{15a}{8}$

684. $-6\frac{2}{5} \div 4$

685. $\left(-15\frac{5}{6}\right) \div \left(-3\frac{1}{6}\right)$

686. $\frac{-6}{\frac{6}{11}}$

687. $\frac{\frac{p}{2}}{\frac{q}{5}}$

688. $\frac{-\frac{4}{15}}{-2\frac{2}{3}}$

689. $\frac{9^2-4^2}{9-4}$

690. $\frac{2}{d} + \frac{9}{d}$

691. $-\frac{3}{13} + \left(-\frac{4}{13}\right)$

692. $-\frac{22}{25} + \frac{9}{40}$

693. $\frac{2}{5} + \left(-\frac{7}{5}\right)$

694. $-\frac{3}{10} + \left(-\frac{5}{8}\right)$

695. $-\frac{3}{4} \div \frac{x}{3}$

696. $\frac{2^3-2^2}{\left(\frac{3}{4}\right)^2}$

697. $\frac{\frac{5}{14}+\frac{1}{8}}{\frac{9}{56}}$

Evaluate.

698. $x + \frac{1}{3}$ when
ⓐ $x = \frac{2}{3}$ ⓑ $x = -\frac{5}{6}$

In the following exercises, solve the equation.

699. $y + \frac{3}{5} = \frac{7}{5}$

700. $a - \frac{3}{10} = -\frac{9}{10}$

701. $f + \left(-\frac{2}{3}\right) = \frac{5}{12}$

702. $\frac{m}{-2} = -16$

703. $-\frac{2}{3}c = 18$

704. Translate and solve: The quotient of p and -4 is -8. Solve for p.